

ERP TEAM COLLABORATION PROJECT

General Objective

The class will develop **one integrated ERP system** divided into **5 major modules**. Each team is responsible for one module, but **all teams must continuously collaborate** because the modules must exchange data and work together as one system.

Important Rule: No team should design its module as a standalone system. Every module must connect to at least **two other teams' modules**.

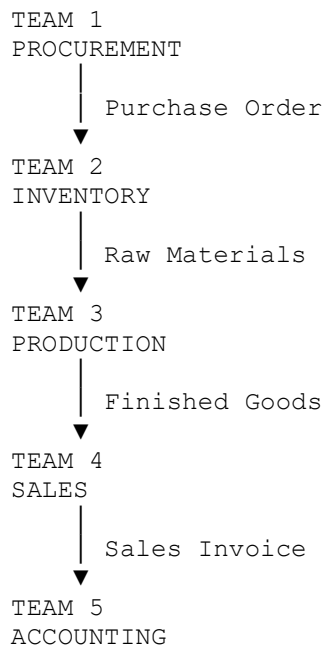
5 DEVELOPMENT TEAMS

Team	Main Responsibility	Primary Module
Team 1	Procurement Team	Procurement & Purchasing
Team 2	Inventory Team	Inventory & Warehouse
Team 3	Production Team	Production & Manufacturing
Team 4	Sales Team	Sales & Customer Management
Team 5	Finance & HR Team	Accounting, Payroll & HRIS

COLLABORATION REQUIREMENT

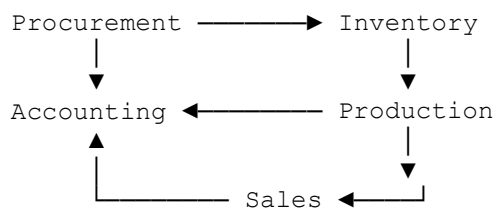
The ERP must work as **ONE SYSTEM**.

Example Integration



But collaboration should **not stop there**.

For example:



TEAM INSTRUCTIONS

TEAM 1 — PROCUREMENT

Responsibilities

Develop the Procurement & Purchasing module.

Required Features

- Supplier Management
- Purchase Request
- Request for Quotation
- Quotation
- Purchase Order
- Goods Receiving
- Purchase Invoice
- Supplier Records
- Purchase Reports

Must Collaborate With

Team 2 – Inventory

When a purchase is received:

```
Purchase Order
    ↓
Goods Receipt
    ↓
Inventory Stock IN
```

Team 5 – Accounting

When a purchase invoice is created:

```
Purchase Invoice
    ↓
Accounts Payable
    ↓
Accounting Journal
```

TEAM 2 — INVENTORY

Responsibilities

Develop Inventory & Warehouse Management.

Required Features

- Item Master
- Raw Materials
- Finished Goods
- Warehouse
- Stock In
- Stock Out
- Stock Transfer
- Stock Adjustment
- Stock Card
- Stock Movement
- Reorder Level
- Low Stock Alert
- Inventory Reports

Must Collaborate With

Team 1 – Procurement

Receiving materials must automatically increase inventory.

Team 3 – Production

Production must be able to request/consume raw materials.

Example:

```
Production Order
  ↓
Material Request
  ↓
Inventory
  ↓
Stock OUT
```

Team 4 – Sales

Delivery of products must reduce inventory.

Sales Order
↓
Delivery
↓
Inventory Stock OUT

TEAM 3 — PRODUCTION

Responsibilities

Develop Production & Manufacturing.

Required Features

- Product Master
- Bill of Materials (BOM)
- Production Order
- Production Planning
- Material Requirement
- Material Issuance
- Production Monitoring
- Finished Goods
- Production Costing
- Production Reports

Must Collaborate With

Team 2 – Inventory

Production consumes raw materials.

Raw Material
↓
Material Issuance
↓
Production
↓
Finished Goods

Finished goods must automatically be added to inventory.

Team 5 – Accounting

Production cost must be available for costing/accounting.

TEAM 4 — SALES

Responsibilities

Develop Sales & Customer Management.

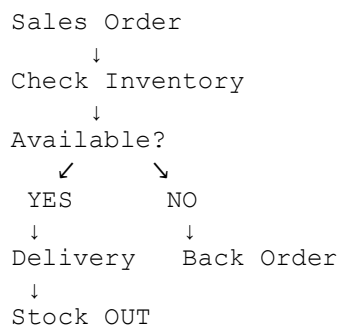
Required Features

- Customer Management
- Sales Quotation
- Sales Order
- Delivery
- Sales Invoice
- Accounts Receivable
- Collection
- Sales Return
- Customer Balance
- Sales Reports

Must Collaborate With

Team 2 – Inventory

Before delivery, the system should check available stock.



Team 5 – Accounting

Sales invoices and collections must affect accounting.

Sales Invoice
↓
Accounts Receivable
↓
Collection
↓
Accounting

TEAM 5 — FINANCE & HR

Responsibilities

Develop:

A. Accounting

- Chart of Accounts
- General Ledger
- Journal Entries
- Accounts Payable
- Accounts Receivable
- Cash/Bank
- Expenses
- Financial Reports
- Trial Balance
- Income Statement
- Balance Sheet

B. HRIS

- Employee Master
- Attendance
- Leave
- Payroll
- SSS
- PhilHealth
- Pag-IBIG
- 13th Month Pay

Must Collaborate With

Team 1 – Procurement

Purchases → AP → Accounting

Team 4 – Sales

Sales → AR → Accounting

Team 3 – Production

Production costs → Cost Accounting

MANDATORY COLLABORATION RULES

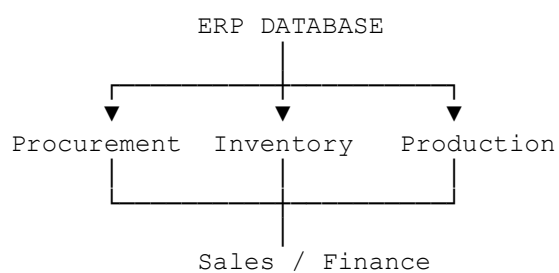
Rule 1 — Shared Database

All teams must use **ONE database**.

Not allowed:

Team 1 Database
Team 2 Database
Team 3 Database

Required:



Rule 2 — Shared Database Design

Before coding, all teams must agree on:

- Table names
- Primary Keys
- Foreign Keys
- Data types
- Naming conventions
- Status values
- User roles
- Transaction numbering

Example:

```
suppliers
supplier_id
supplier_name

purchase_orders
po_id
supplier_id

goods_receipts
gr_id
po_id

stock_movements
movement_id
item_id
reference_id
```

Rule 3 — Every Team Must Have Dependencies

Each team must identify:

INPUT

Where does your module get information?

PROCESS

What does your module do?

OUTPUT

Which team uses your information?

Example:

Procurement

INPUT
Supplier + Items

↓

PROCESS
Purchase Order

↓

OUTPUT
Goods Receipt
↓
Inventory Team

REQUIRED COLLABORATION MEETINGS

Each team must conduct regular coordination meetings.

Meeting 1 — System Planning

All 5 teams agree on:

- ERP objectives
- Modules
- Database
- Users
- Workflow
- UI standards

Meeting 2 — Database Integration

Teams finalize:

- ERD
- Tables
- Relationships
- Foreign Keys

Meeting 3 — API/Data Integration

Teams discuss how modules exchange data.

Meeting 4 — Integration Testing

Teams test their modules together.

Meeting 5 — Final ERP Testing

All teams perform an end-to-end transaction.

REQUIRED END-TO-END TEST

The final ERP must demonstrate this scenario:

```
1. Procurement creates Purchase Request
    ↓
2. Purchase Order is created
    ↓
3. Inventory receives materials
    ↓
4. Production uses raw materials
    ↓
5. Finished Goods enter Inventory
    ↓
6. Sales creates Sales Order
    ↓
7. Inventory checks stock
    ↓
8. Delivery is created
    ↓
9. Sales Invoice is created
    ↓
10. Accounting records AR
    ↓
11. Customer pays
    ↓
12. Accounting records Collection
```

If one team's module breaks the workflow, the ERP is considered incomplete.

PROJECT REQUIREMENTS

Every team must submit:

1. Documentation

- Module description
- Objectives
- Functional requirements
- Non-functional requirements
- Use Case Diagram
- Flowchart
- ERD
- Data Dictionary

2. System

- Login
- Dashboard
- CRUD operations
- Search
- Filtering
- Reports
- Validation
- Error handling
- User permissions
- Audit trail

3. Integration

Each team must demonstrate:

- **At least 2 working integrations**
- Shared database
- Foreign-key relationships
- Data flowing between modules
- No duplicate master data

4. Testing

Submit:

- Test cases
- Test results

- Integration testing
- Bug/issue report
- Final system testing

IMPORTANT TEAM RULE

"ONE ERP, FIVE TEAMS, ONE GOAL."

Students should understand that **they are not creating five separate projects.**

They are creating:

ONE ERP SYSTEM developed by five collaborating teams.

Each team's success depends on the success of the other teams.